Ushtrime ND10a

2.1.8 *Thupra metalike në temperaturën $t_1 = 20^0 C$ ka gjatësinë $l_1 = 20cm$, e cila gjatë ndryshimit të temperaturës për $\Delta t = 30^0 C$ zgjatet për $\Delta l = 0{,}138mm$. Sa është koeficienti termik i bymimit linear?*

Zgjidhje:

Të dhënat:

$t_1 = 20°c$

$l_1 = 20cm = 20 \cdot 10^{-2} m$

$\Delta t = 30°c$

$\Delta l = 0{,}138mm = 0{,}138 \cdot 10^{-3} m$

$\alpha = ?$

$\Delta l = l - l_0$ ndryshimi absolut i gjatësisë

$\Delta l_r = \frac{\Delta l}{l_0}$ gjatësia relative.

$$\alpha = \frac{\Delta l_r}{\Delta t} = \frac{\Delta l}{l_0 \cdot \Delta t} = \frac{1}{\Delta t} \cdot \frac{\Delta l}{l_0}$$

$\Delta t = t - \cancel{t_0^\circ} = t$

$$\alpha = \frac{1}{t} \cdot \frac{\Delta l}{l_0} \Big/ \cdot t \cdot l_0$$

$\alpha \cdot t \cdot l_0 = \Delta l$

$\alpha \cdot t \cdot l_0 = l - l_0$

$l = \alpha \cdot t \cdot l_0 + l_0 \rightarrow l = l_0(1 + \alpha t)$ ose

$l = l_0(1 + \alpha \Delta t)$

për rastin nga detyra:

$l_1 = l_0(1 + \alpha t_1) \Rightarrow l_0 = \frac{l_1}{1 + \alpha t_1}$

$l_2 = l_0(1 + \alpha t_2)$

$$l_2 = \frac{l_1(1 + \alpha t_2)}{1 + \alpha t_1} \Big/ \cdot (1 + \alpha t_1)$$

$l_2(1 + \alpha t_1) = l_1(1 + \alpha t_2)$

$l_2 + l_2 \alpha t_1 = l_1 + l_1 \alpha t_2$

$\underbrace{l_2 - l_1}_{\Delta l} = l_1 \alpha t_2 - l_2 \alpha t_1$

$\Delta l = \alpha(l_1 t_2 - l_2 t_1) \Big/ : (l_1 t_2 - l_2 t_1)$

$$\alpha = \frac{\Delta l}{l_1 t_2 - l_2 t_1}$$

$\Delta t = t_2 - t_1 \Rightarrow t_2 = \Delta t + t_1$

$\Delta l = l_2 - l_1 \Rightarrow l_2 = \Delta l + l_1$

$$\alpha = \frac{\Delta l}{l_1(\Delta t + t_1) - t_1(\Delta l + l_1)}$$

$$\alpha = \frac{\Delta l}{l_1 \Delta t + \cancel{l_1 t_1} - t_1 \Delta l - \cancel{t_1 l_1}}$$

$$\alpha = \frac{\Delta l}{l_1 \Delta t - t_1 \Delta l}$$

$$\alpha = \frac{0{,}138 \cdot 10^{-3} m}{20 \cdot 10^{-2} m \cdot 30°c - 20°c \cdot 0{,}138 \cdot 10^{-3} m}$$

$$\alpha = \frac{0{,}138 \cdot 10^{-3} m}{600 \cdot 10^{-2} m°c - 2{,}76 \cdot 10^{-3} m°c}$$

$$\alpha = \frac{0{,}138 \cdot 10^{-3} m}{6000 \cdot 10^{-3} m°c - 2{,}76 \cdot 10^{-3} m°c} = \frac{0{,}138 \cdot \cancel{10^{-3}} m}{(6000 - 2{,}76) \cdot \cancel{10^{-3}} m°c}$$

$$\alpha = \frac{0{,}138}{5997{,}24 °c} = \frac{0{,}138}{5{,}99724} \cdot 10^{-3} \frac{1}{°c} = 0{,}02301 \cdot 10^{-3} \frac{1}{°c}$$

$$\boxed{\alpha = 2{,}301 \cdot 10^{-5} \frac{1}{°c}}$$